

Visual Data Science

Discover

Describe your topic

This project examines **global investment in renewable energy research**, development, and demonstration (RD&D). The primary objective is to analyse national RD&D expenditures and **identify leading countries in this area**, normalised by their respective gross domestic product (GDP).

In the first stage, the study will compare the scale of RD&D investments across countries to determine which nations are at the **forefront of renewable energy innovation**. Subsequently, the analysis will investigate how countries allocate their RD&D budgets across different renewable energy technologies, **exploring both current priorities and historical trends**

The project will also evaluate the broader impact of RD&D investments by assessing their correlation with key environmental and energy performance indicators. In particular, it will examine the **relationship between investment levels and (i) carbon dioxide (CO₂) emissions, and (ii) the share of renewable energy in total energy generation**. Data will be visualised to illustrate causal links and highlight the potential benefits of sustained investment in renewable energy RD&D.

Describe the dataset you selected

For this project, I utilise multiple datasets related to renewable energy, including indicators such as RD&D investments, energy generation, carbon emissions, and historical GDP by country. **All datasets are provided in CSV format** and are structured by country and year, enabling both time-series and cross-country analysis.

The integration of these datasets enables an analysis of how national investments in renewable energy evolve over time and how these investments translate into increased renewable energy generation. Furthermore, it allows for the assessment of their environmental impact by examining the correlation between RD&D investment levels and CO₂ emissions reduction.

All datasets, along with detailed documentation of their structure and sources, are available in my GitHub repository:

 <https://github.com/mozi30/VDS-RenewableEnergy.git>